ARITHMETIC ASSIGNMENT 7

TIME & WORK

Q156. Three pipes, A, B and C are opened to fill an empty tank such that A and B cam fill the tank alone in 36 minutes and 45 minutes respectively and C can empty it in 30 minutes. After 6 minutes the emptying pipe is closed. In how many minutes the tank will be filled ?

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Q157. Pipes A and B can fill a tank in 5 and 6 hours respectively. Pipe C can empty it in 12 hours. If all the three pipes are opened together, then the tank will be filled in.

ELITES GRID
@cat prep

Q158. 3 pipes can fill a reservoir in 10,15 and 20 hours respectively . If three taps are opened one after another in the given order . with a certain fixed time gap between them . The reservoir fills in 5 hours . find the time gap:

ELITES GRID
@cat prep

Q159. A tank has four inlet pipes such that each inlet pipe while working independently can fill the tank in 4 hours. The tank also has two outlet pipes such that each outlet pipe while working independently can empty the tank in 3 hours. If all the six pipes are opened simultaneously, then in how much time will the tank get filled completely?

(a) 2.0 hours

(b) 3.0 hours

(c) 2.5 hours

(d) 7.0 hours

ELITES GRID
@cat prep

Q160. It takes 30 hours for an inlet pipe to fill an empty tank completely. When 5 identical inlet pipes and 4 identical outlet pipes operate together, the same empty tank get filled completely in 10 hours.
How much time (in hours) will an outlet pipe take to empty the same tank when it’s filled upto half its volume?
a) 15 (b) 20 (c) 24 (d) 30

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Q161.(CAT 2017)A tank has an inlet pipe and an outlet pipe. If the outlet pipe is closed then the inlet pipe fills the empty tank in 8 hours. If the outlet pipe is open then the inlet pipe fills the empty tank in 10 hours. If only the outlet pipe is open then in how many hours the full tank becomes half-full?

A. 20

B. 30

C. 40

D.45

ELITES GRID
@cat prep

Q162. (CAT 2018) -A tank is fitted with pipes, some filling it and the rest draining it. All filling pipes fill at the same rate, and all draining pipes drain at the same rate. The empty tank gets completely filled in 6 hours when 6 filling and 5 draining pipes are on, but this time becomes 60 hours when 5 filling and 6 draining pipes are on. In how many hours will the empty tank get completely filled when one draining and two filling pipes are on .

ELITES GRID
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Q163. (CAT 2018)- A water tank has inlets of two types A and B. All inlets of type A when open, bring in water at the same rate. All inlets of type B, when open, bring in water at the same rate. The empty tank is completely filled in 30 minutes if 10 inlets of type A and 45 inlets of type B are open, and in 1 hour if 8 inlets of type A and 18 inlets of type B are open.
In how many minutes will the empty tank get completely filled if 7 inlets of type A and 27 inlets of type B are open?

ELITES GRID
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Q164. A and B working together can finish a job in T days. If A works alone and completes the job, he will take T+5 days. If B works alone and completes the same job, he will take T+45 days. What is T?

ELITES GRID
@cat prep

Q165. If Amit is 33.33% more efficient than Sumit, Amit worked for x days and Sumit do the remaining work in x+3 days. The work ratio of Amit and Sumit is 2 : 3 respectively. Find the value of x? (Q number 165 to Q 174 Video Will be updated on Monday 11:00 PM)

ELITES GRID
@cat prep

Q166. If A and B work together, they will complete a job in 7.5 days. However, if A works alone and completes half the job and then B takes over and completes the remaining half alone, they will be able to complete the job in 20 days. How long will B alone take to do the job if A is more efficient than B?

ELITES GRID
@cat prep

Q167. Abdul, Bimal, Charlie and Dilbar can finish a task in 10, 12, 15 and 18 days respectively. They can either choose to work or remain absent on a particular day.
If 50 percent of the total work gets completed after 3 days, then, which of the following options is possible?

A. Each of them worked for exactly 2 days.

B. Bimal and Dilbar worked for 1 day each, Charlie worked for 2 days and Abdul worked for all 3 days.

C. Abdul and Charlie worked for 2 days each, Dilbar worked for 1 day and Bimal worked for all 3 days.

D. Abdul and Dilbar worked for 2 days each, Charlie worked for 1 day and Bimal worked for all 3 days.

E. Abdul and Charlie worked for 1 day each, Bimal worked for 2 days and Dilbar worked for all 3 days

ELITES GRID
@cat prep

Q168. When working alone X, Y and Z can complete a piece of work in 8, 12 and 30 days respectively. At the most only two people can work on each day and nobody works for more than two consecutive days. What is the minimum number of days that they will take to finish the work.

1) 4^(4/29)

2) 6^(4/25)

3) 6^(6/29)

4) 12^(4/25)

ELITES GRID
@cat prep

Q169. Aman, Baman and Chaman can finish a job working alone in 15, 20 and 25 days respectively. However, while working with somebody the efficiency of Aman, Baman and Chaman reduces by 30%, 20% and 50% respectively. If none of them is allowed to work for three consecutive days, then what is the maximum possible percentage of the job that they can complete in four days

ELITES GRID
@cat prep

Q170. A , B , C can do a piece of work in 20, 30 and 60 days respectively. In how many days can A do the work if he is assisted by B and C on every 3rd day.

ELITES GRID
@cat prep

Q171. P and Q working together can do a piece of work in 10 days whereas Q and R working together can do the same work in 12 days. All three work together to do a job for which they are paid Rs. 300. If Q's share is Rs. 140, then what is P's share?

(a) Rs. 100

(b) Rs. 60

(c) Rs. 80

(d) Cannot be determined

ELITES GRID
@cat prep

Q172. A group of 10 men can do a job in 15 days. The group starts working and after every three days they are joined by two more men. On which day does the job get completed?

(a)10th

(b)11th

(c)12th

(d)13th.

ELITES GRID
@cat prep

Q173. **With P starting the work, working on alternate days, P and Q can finish a work in 17 days. If Q works on the first day and Q and P work alternately, the work is finished in 53/3 days. How many days Q will take individually to execute the work?**

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@cat prep

Q174. **8 workers working for 8 days make 8 motor cycles. Then how many motor cycles would be made by 4 workers if they work for 4 days? Assume equal efficiency of each worker.**

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Ans 156- 24

Ans 157- 60/17 hours

Ans 158- 1/2 hours

Ans 159- 3 hours

Ans 160- 30

Ans 161- 20

Ans 162- 10

Ans 163- 48

Ans 164- 15

Ans 165- 3

Ans 166- 30 days

Ans 167- E

Ans 168- 2nd option

Ans 169- 34%

Ans 170- 15

Ans 171 - Rs 100

Ans 172- 12th

Ans 173- 35 days

Ans 174- 2

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ARITHMETIC ASSIGNMENT 6

Time & Work

Q144. **A person can complete a job in 120 days. He works alone on Day 1. On Day 2, he is joined by another person who also can complete the job in exactly 120 days. On Day 3, they are joined by another person of equal efficiency. Like this, everyday a new person with the same efficiency joins the work. How many days are required to complete the job?**

ELITES GRID
@cat prep

Q145. At their usual efficiency levels, A and B together finish a task in 12 days. If A had worked half as efficiently as she usually does, and B had worked thrice as efficiently as he usually does, the task would have been completed in 9 days. How many days would A take to finish the task if she works alone at her usual efficiency?

ELITES GRID
@cat prep

Q146. One day, Rahul started a work at 9 AM and Gautam joined him two hours later. They then worked together and completed the work at 5 PM the same day. If both had started at 9 AM and worked together, the work would have been completed 30 minutes earlier. Working alone, the time Rahul would have taken, in hours, to complete the work is

ELITES GRID
@cat prep

Q147. Ramesh and Ganesh can together complete a work in 16 days. After seven days of working together, Ramesh got sick and his efficiency fell by 30%. As a result, they completed the work in 17 days instead of 16 days. If Ganesh had worked alone after Ramesh got sick, in how many days would he have completed the remaining work?

ELITES GRID
@cat prep

Q148. Three men and eight machines can finish a job in half the time taken by three machines and eight men to finish the same job. If two machines can finish the job in 13 days, then how many men can finish the job in 13 days?

ELITES GRID
@cat prep

Q149. When they work alone, B needs 25% more time to finish a job than A does. They two finish the job in 13 days in the following manner: A works alone till half the job is done, then A and B work together for four days, and finally B works alone to complete the remaining 5% of the job. In how many days can B alone finish the entire job?

ELITES GRID
@cat prep

Q150. **The amount Neeta and Geeta together earn in a day equals what Sita alone earns in 6 days. The amount Sita and Neeta together earn in a day equals what Geeta alone earns in 2 days. The ratio of the daily earnings of the one who earns the most to that of the one who earns the least is**

A) 11:3

B) 11:7

C) 1:2

D) 3:7

ELITES GRID
@cat prep

Q151. Working alone, the times taken by Anu, Tanu and Manu to complete any job are in the ratio 5 : 8 : 10. They accept a job which they can finish in 4 days if they all work together for 8 hours per day. However, Anu and Tanu work together for the first 6 days, working 6 hours 40 minutes per day. Then, the number of hours that Manu will take to complete the remaining job working alone is

ELITES GRID
@cat prep

Q152. **Bob can finish a job in 40 days, if he works alone. Alex is twice as fast as Bob and thrice as fast as Cole in the same job. Suppose Alex and Bob work together on the first day, Bob and Cole work together on the second day, Cole and Alex work together on the third day, and then, they continue the work by repeating this three-day roster, with Alex and Bob working together on the fourth day, and so on. Then, the total number of days Alex would have worked when the job gets finished, is**

ELITES GRID
@cat prep

Q153. **Gautam and Suhani, working together, can finish a job in 20 days. If Gautam does only 60% of his usual work on a day, Suhani must do 150% of her usual work on that day to exactly make up for it. Then, the number of days required by the faster worker to complete the job working alone is**

ELITES GRID
@cat prep

Q154. John takes twice as much time as Jack to finish a job. Jack and Jim together take one-thirds of the time to finish the job than John takes working alone. Moreover, in order to finish the job, John takes three days more than that taken by three of them working together. In how many days will Jim finish the job working alone?

ELITES GRID
@cat prep

Q155. Twelve men can complete a work in 8 days. Three days after the work 3 more men joined. In how many days will all of them together complete the remaining work?

ELITES GRID
@cat prep

ANS KEY

Ans 144) 15

Ans 145) 18

Ans 146) 10

Ans 147) 13.5

Ans 148) 13

Ans 149) 20

Ans 150) 11:3

Ans 151) 6 hours

Ans 152) 11

Ans 153) 36

Ans 154) 4

Ans 155) 4

ARITHMETIC ASSIGNMENT 5
Time & Work

Q116. A takes 10 days to complete a work and B takes 25 days to complete a work.
In how many days can A and B together complete the work.

ELITES GRID
@cat prep

Q117. Hunny Malhotra can do a work in 10 days . After working for 4 days he was joined by Abhinav Tyagi . If they complete the remaining work in 2 more days. In how many days Abhinav alone can complete the work

ELITES GRID
@cat prep

Q118. Aashna, Risitha, Vineet can do a work in 15 days. After all three worked for 2days, Aashna left, Risitha and Vineet worked for 10 more days and then Risitha left .. Vineet worked for another 40 days and completed the work. In how many days Aashna alone complete the work If Vineet alone can complete it in 75 days.

ELITES GRID
@cat prep

Q119. A and B working alone can do a certain piece of work in 40 days and 120 days respectively. A and B start working together on that work but after having worked for 10 days, additional work came in. The additional work is such that B would take 40 days working alone to complete it . In order to cope with the additional load, C is induced and the remaining work is finished in 5 days. Find the number of days that C alone would take to complete the original work.

ELITES GRID
@cat prep

Q120. A can do a piece of work in 20 days and B in 30 days. They work together for 7 days and then both leave the work. Then C alone finishes the remaining work in 10 days. In how many days will C finish the complete work?

ELITES GRID
@cat prep

Q121. A, B, and C can do a work separately in 16, 32 and 48 days respectively. They started the work together but B left 8 days and C six days before completion of the work. What was the duration of completion of the work?

ELITES GRID
@cat prep

Q122. Ram and Laxman are assigned the task to build a wall. Ram can complete the task in 30 days, while Laxman can complete the same task in 60 days. When they work together, their efficiency drops to 80%. How long will they take to complete the task working together?

a. 15 days

b. 20 days

c. 25 days

d. 32 days

ELITES GRID
@cat prep

Q123. Gee-k can do a piece of work in 20 days and Rishi can do the same in 30 days. They finished the work with the help of Prateek in 8 days. If they earned a total of 5550 Rs . What is share of Prateek.

ELITES GRID
@cat prep

Q124. Anil alone can do a job in 20 days while Sunil alone can do it in 40 days. Anil starts the job, and after 3 days, Sunil joins him. Again, after a few more days, Bimal joins them and they together finish the job. If Bimal has done 10% of the job, then in how many days was the job done ? (CAT 2019)

ELITES GRID
@cat prep

Q125. Amal can complete a job in 10 days and Bimal can complete it in 8 days. Amal, Bimal and Kamal together complete the job in 4 days and are paid a total amount of Rs 1000 as remuneration. If this amount is shared by them in proportion to their work, then Kamal's share, in rupees , is (CAT 2017)

A) 100
B) 200
C) 300
D) 400

ELITES GRID
@cat prep

Q126. Two painters, A and B are under contract to paint a certain wall every day. Their rate of painting is constant and never varies, even from day to day. On day 1, they notice that they can paint the wall together in 'x' minutes. The next day, A does not turn up and B works alone. B notices that he takes 5 minutes more than they had taken the previous day to paint the wall. On day 3, B does not turn up. A calculates the time he took to paint the wall alone and tells B that he had taken 40 minutes more than B had taken on day 2. What is the value of x ?

ELITES GRID
@cat prep

Q127. 2 men and 4 boys can complete a piece of work in 8 days . 4 men and 2 boys can complete the same work in 6 days . How much time does a man and a boy working together require to complete the work

ELITES GRID
@cat prep

Q128. A piece of work can be done by 11 men and 16 boys in 2 days. The same work can be done by 5 men and 11 boys in 4 days. In how many days can 1 man and 4 boys complete the same work

ELITES GRID
@cat prep

Q129. Humans and robots can both perform a job but at different efficiencies. Fifteen humans and five robots working together take thirty days to finish the job, whereas five humans and fifteen robots working together take sixty days to finish it. How many days will fifteen humans working together (without any robot) take to finish it? (CAT 2018)

ELITES GRID
@cat prep

Q130. 10 men and 12 children complete a certain piece of work in 10 days. Each child takes thrice the time taken by a man to complete the work. The time taken by 12 men to finish the same work is

ELITES GRID
@cat prep

Q131. 5 men and 8 women can do a job in 8 days, while 4 men and 6 women can do it in 10 days. How many days will 10 women take to finish the job?

[A] 10 days

[B] 40 days

[C] 20 days

[D] None of these

ELITES GRID
@cat prep

Q132. Anil can paint a house in 60 days while Bimal can paint it in 84 days. Anil starts painting and after 10 days, Bimal and Charu join him. Together, they complete the painting in 14 more days. If they are paid a total of ₹ 21000 for the job, then the share of Charu, in INR, proportionate to the work done by him, is (CAT 2021)

A) 9000

B) 9100

C) 9200

D) 9150

ELITES GRID
@cat prep

Q133. Amar, Akbar and Anthony are working on a project. Working together Amar and Akbar can complete the project in 1 year, Akbar and Anthony can complete in 16 months, Anthony and Amar can complete in 2 years. If the person who is neither the fastest nor the slowest works alone, the time in months he will take to complete the project is (CAT 2021)

ELITES GRID
@cat prep

Q134. Anil can paint a house in 12 days while Barun can paint it in 16 days. Anil, Barun, and Chandu undertake to paint the house for ₹ 24000 and the three of them together complete the painting in 6 days. If Chandu is paid in proportion to the work done by him, then the amount in INR received by

ELITES GRID
@cat prep

Q135. Working alone, Ram can erect the boundary wall of a land in 16 days but Samar can do the same work in 12 days. Ram and Samar are put to work on that contract on alternate days. In how many days will the boundary wall be ready, if Ram works on first day?

A. 55/4 days B. 10 Days C. 12 days D. 51/4

ELITES GRID
@cat prep

Q136. A and B, working together can build a wall, 221 m long in 100/9 days. If they work on alternate days, with A starting the work, it takes 89/4 days to build the same wall. If A and B work together and build a similar wall but of twice the length and earn a total of Rs. 1800 for it . Then B's Share of the earnings will be

ELITES GRID
@cat prep

Q137. **Anu, Vinu and Manu can complete a work alone in 15 days, 12 days and 20 days, respectively. Vinu works everyday. Anu works only on alternate days starting from the first day while Manu works only on alternate days starting from the second day. Then, the number of days needed to complete the work is**

A) 5

B) 6

C) 7

D) 8

ELITES GRID
@cat prep

Q138. If 6 men working 5 days for 8 hours each can finish a task. How many men would be needed for 4 days for 6 hours each to finish half the task.

ELITES GRID
@cat prep

Q139. 12 examiners (men) work 16 hours a day to check 24000 answer sheets in 18 days. Now, 24 examiners would work how many hours per day to check 36000 answer sheets in 36 days?

ELITES GRID
@cat prep

Q140. A contractor agreed to finish a piece of work in 150 days. He employed 75 men and made them work for 8 hours per day. However, after 90 days, he realized that only (2/7 th) of the work was completed. How many more men would he need to employ to complete the work on time if he intended to make everybody work for 10 hours per day henceforth?

(a) 75 (b) 100 (c) 150 (d) 225

ELITES GRID
@cat prep

Q141. If 15 men or 24 women or 36 boys can complete a piece of work in 12 days by working 8 hours a day, then how many men must be required to work with 12 women and 6 boys to complete another piece of work 9/4 times the previous piece of work in 30 days by working 6 hours a day

ELITES GRID
@cat prep

Q142. **A contractor agreed to construct a 6 km road in 200 days. He employed 140 persons for the work. After 60 days, he realized that only 1.5 km road has been completed. How many additional people would he need to employ in order to finish the work exactly on time?**

ELITES GRID
@cat prep

Q143. A group of people worked on a project. They finished 35% of the project by working 7 hours a day for 10 days. Thereafter, 10 people left the group and the remaining people finished the rest of the project in 14 days by working 10 hours a day. Then the value of is

ELITES GRID
@cat prep

ANS KEY

Ans 116) 50/7 days

Ans 117) 5

Ans 118) 30

Ans 119) 6

Ans 120) 24

Ans 121) 12

Ans 122) 25

Ans 123) 1850 Rs

Ans 124) 13 days

Ans 125) 100

Ans 126) 15

Ans 127) 144/7

Ans 128) 16.4 days

Ans 129) 32

Ans 130) 11.66 days

ELITES GRID
@cat prep

ANS KEY

Ans 131) None of these

Ans 132) 9100

Ans 133) 32 months

Ans 134) Rs 3000

Ans 135) 55/4 days

Ans 136) 1000 Rs

Ans 137) 7

Ans 138) 5

Ans 139) 6

Ans 140) 150

Ans 141) 8

Ans 142) 40

Ans 143) 140

ELITES GRID
@cat prep